Map Localization for ADAS

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Outline

Motivation

A diagram

A plot

Some code

A video

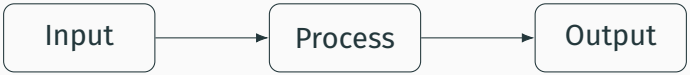
Motivation

Why we care

- One reason
- Another reason
- A third one with emphasis

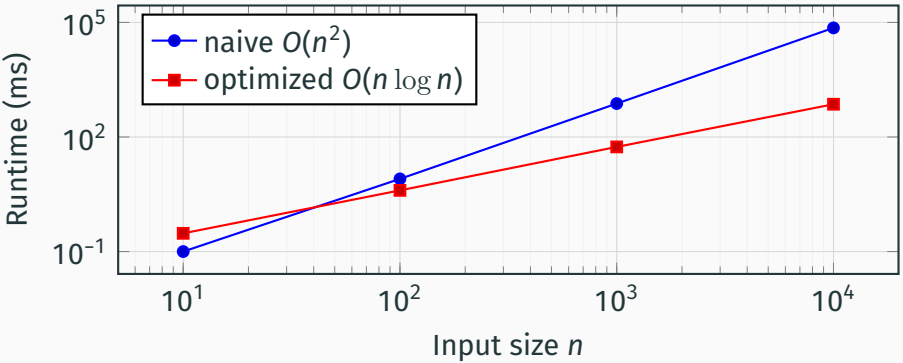
A diagram

TikZ works exactly like under pdflatex



A plot

Benchmark: runtime vs. input size



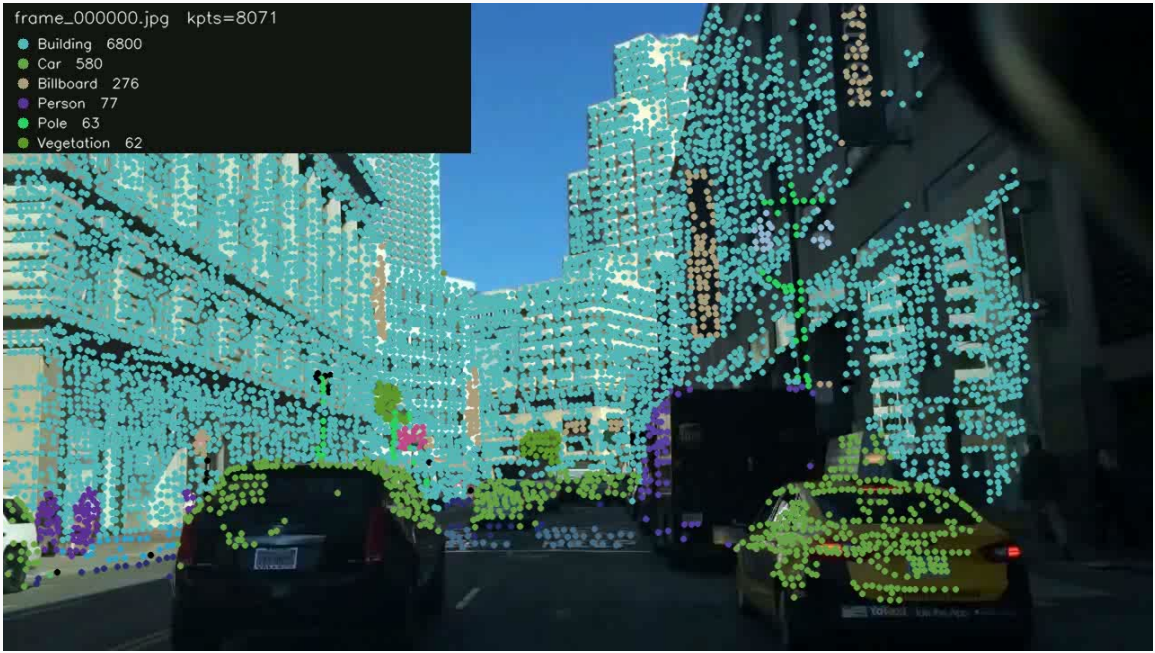
Some code

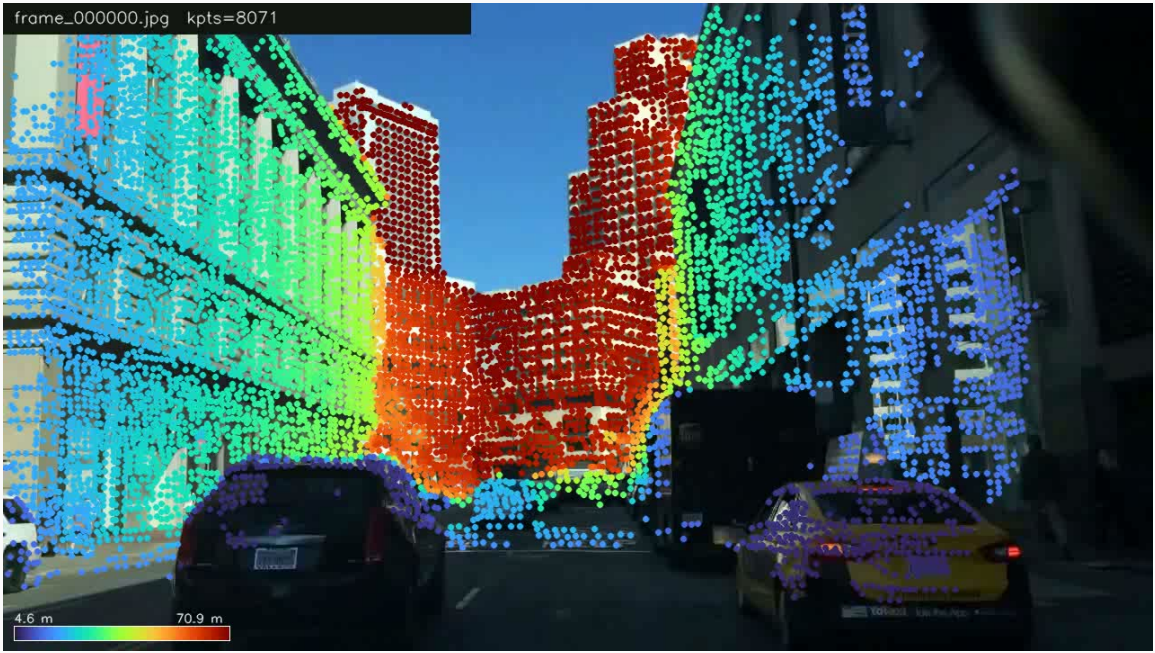
A syntax-highlighted code block

```
def greet(name: str) -> str:
    return f"hello, {name}"
```

A video

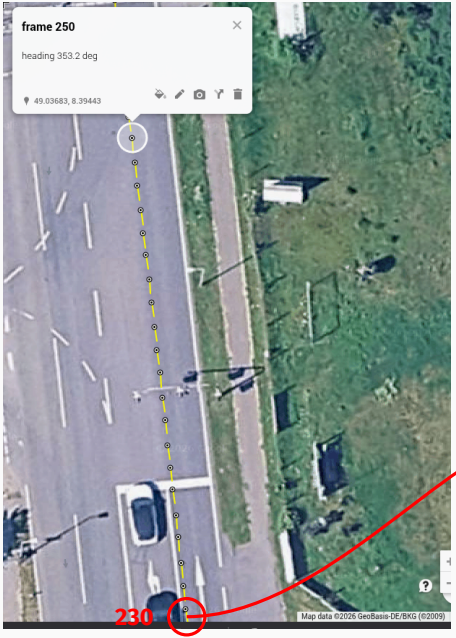

perception cache v1 step=3
frame 000000 / 000099 npz: yes
semantic
instance
mono_depth
depth_conf


frame_000000.jpg kpts=8071
Building 6800
Car 580
Billboard 276
Person 77
Pole 63
Vegetation 62


frame_000000.jpg kpts=8071
4.6 m
70.9 m


frame 000/270 ego-lane=green border=black | yellow=arrow magenta=sign green=light white=guidepost | red=moving orange=pot_dyn (in-box) (poles excluded)
car_pot_dyn
car_DYN
car_DYN
A0
70
70



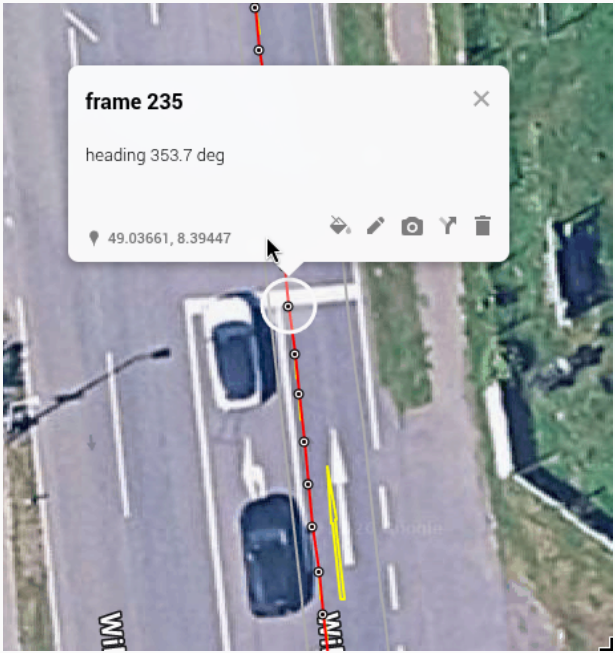


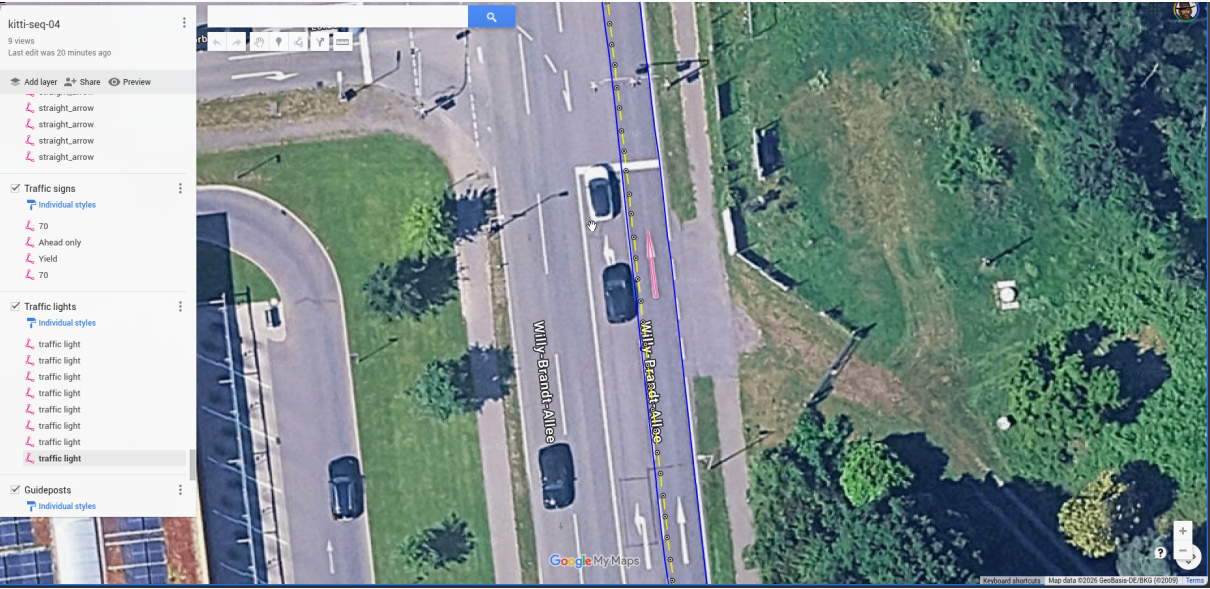
KITTI used a combined GPS/IMU system with **Real-Time Kinematic (RTK) correction**, which provides a localization accuracy/error of approximately **5 cm.** KIT - Karlsruher Institut für Technologie +1

Die Kontinentaldrift in Europa beträgt für den Zeitraum 2011–2026 (15 Jahre) insgesamt etwa **15 cm bis 60 cm**, da sich die eurasische Landmasse je nach Region um **1 cm bis 4 cm pro Jahr** verschiebt. Geo.de +1

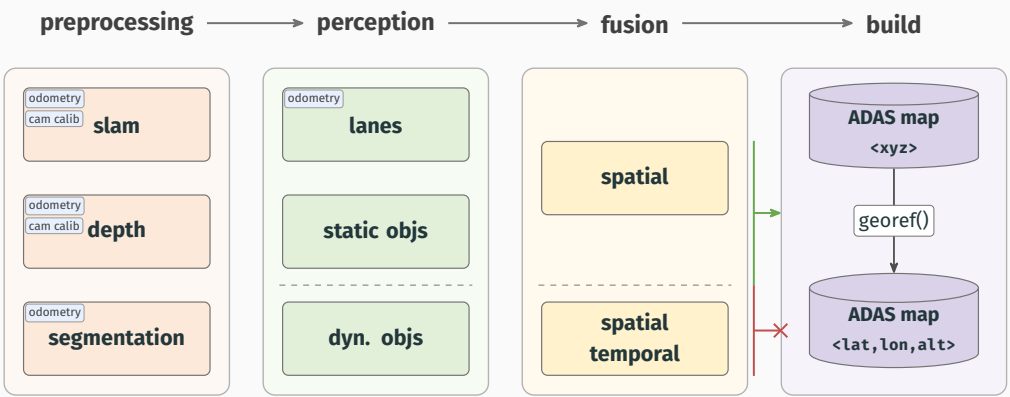
Die GPS-Genauigkeit von Smartphones auf Google Maps liegt im Freien meist bei etwa **5 bis 20 Metern** Google Help. Satellitenbilder bieten in Städten teilweise eine Auflösung von bis zu **50 cm**, die absolute Positionsgenauigkeit der Karten kann durch Verzerrungen aber um **bis zu 10 Meter oder mehr** abweichen. Google Help +2




frame 235
heading 353.7 deg
49.03661, 8.39447


kitti-seq-04
9 views
Last edit was 20 minutes ago
Add layer
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straight_arrow
straight_arrow
straight_arrow
straight_arrow
Traffic signs
Individual styles
70
Ahead only
Yield
70
Traffic lights
Individual styles
traffic light
traffic light
traffic light
traffic light
traffic light
traffic light
traffic light
traffic light
Guideposts
Individual styles
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Willy-Brandt-Allee
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Map Building Workflow



Questions?